

AttnSeq-PPI: Self and Cross Attention-Driven Prediction of sequence-based Protein-Protein Interactions

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Abstract

Protein-protein interactions (PPIs) are a major component of cellular organization, function, and biochemical reactions. Experimental detection of PPIs through high-throughput techniques as well as structure-based docking is costly and time-consuming. Existing computational techniques that use statistical measures or machine-learning algorithms are prone to overfitting and vulnerable to data noise or bias. To overcome these problems, we propose a deep-learning-based PPI prediction model, AttnSeq-PPI which is a combination of self and cross-attention with convolution operation. Protein sequences were embedded to vectors using the word2vec algorithm to generate a feature matrix. The attention mechanism captures relationships between distant elements in a sequence and understands complex patterns and dependencies. The 1-D convolution operation is used to extract the local features of the sequence, reducing the dimension of the vector and computational cost. Our proposed model was trained using intra-species human and multi-species datasets, validation was performed using four independent species and PPI network datasets. AttnSeq-PPI is very effective in performing the prediction and also in generalization ability, it outperformed the existing state-of-the-art models. The accuracy of our proposed model is 98.81% and recall is 98.67 % for the human dataset and can predict unknown protein pairs with fewer false negatives with higher precision.

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